

On the normalization of Eqs. (7)–(9) of Hsu & Lai,
Angle-Resolved Spectroscopy of Parametric Fluorescence
 (arXiv:1302.6226) — angle conventions and the genuine misprints

The paper’s master formula for the angle-resolved Type-I SPDC photon flux (used to compute Fig. 6) is Eq. (9). An earlier version of this note claimed that the printed Eq. (9) was $2\pi n_s^2 \approx 17\times$ too low, attributing the deficit to a dropped coefficient in the angular Jacobian. That claim was **wrong**: it rested on reading θ_s as the *internal* (in-crystal) angle. The paper’s Fig. 2 defines the angle conventions explicitly — “The angles θ'_s and θ_s are, respectively, internal and external angles formed by the signal and pump wave vectors” — and with θ_s the *external* (laboratory) angle the printed Eq. (9) is exactly correct as a flux density per unit azimuth. Its only blemish is a dropped $d\phi$ symbol. The genuine misprints are confined to the main-text Eq. (7): the prefactor constant ($2\pi^4$ vs the appendix’s correct $8\pi^4$) and the sign of the Gaussian exponent.

The relevant printed equations

Eq. (7), main text — flux per unit frequency:

$$N_s(\omega_s, \kappa_s) = \frac{\hbar d_{\text{eff}}^2 \omega_s \omega_i \omega_p L^2 N_p}{2\pi^4 c^3 \varepsilon_0 n_s n_i n_p} d\omega_s d^2\kappa_s \int d^2\xi e^{+\frac{1}{2}\xi^2} \text{sinc}^2\left(\frac{1}{2}L \Delta k_z\right). \quad (7)$$

(Both bold-faced items are misprints; see below. The exponent must be $-\frac{1}{2}\xi^2$, as the appendix has it.)

Appendix intermediate (after relabeling $\xi \rightarrow \xi_i$ and the saddle-point integral over the *relative* azimuth between κ_s and κ_i , which supplies the $\sqrt{2\pi}$):

$$N_s = \underbrace{\frac{\hbar d_{\text{eff}}^2 \omega_s \omega_i \omega_p L^2 N_p \sqrt{2\pi}}{8\pi^4 c^3 \varepsilon_0 n_s n_i n_p}}_{A_0} d\omega_s \underbrace{\kappa_s d\kappa_s d\phi}_{=d^2\kappa_s} \int d\xi_i e^{-\frac{1}{2}(\xi_s - \xi_i)^2} \text{sinc}^2\left(\frac{1}{2}L \Delta k_z\right). \quad (1)$$

Note the signal azimuth $d\phi$ is still present here.

Eq. (9), as printed (target: variables λ_s, θ_s):

$$N_s(\lambda_s, \theta_s) = 2\pi\sqrt{2\pi} \frac{\hbar d_{\text{eff}}^2 \omega_p L^2 N_p}{\varepsilon_0 n_s n_i n_p \lambda_s^5 \lambda_i} \sin(2\theta_s) d\lambda_s d\theta_s \int d\xi_i e^{-\frac{1}{2}(\xi_s - \xi_i)^2} \text{sinc}^2\left(\frac{1}{2}L \Delta k_z\right). \quad (9)$$

The angle convention: θ_s is the laboratory angle

The paper primes all in-medium quantities: Eq. (3) and Eq. (6) use the internal angle θ'_s , and Eq. (6) and the appendix define the in-medium wavenumbers as $k'_s = n_s \omega_s / c$, etc. Accordingly, the appendix's substitution " $\kappa_s = k_s \sin \theta_s$ " (unprimed k_s , unprimed θ_s) means the *vacuum* wavenumber $k_s = k_0 = 2\pi/\lambda_s$ with the *external* angle. This is also the physically correct statement of the conserved transverse momentum: κ is continuous across the flat exit face (Snell), so

$$\kappa_s = k_0 \sin \theta_s^{\text{ext}} = k'_s \sin \theta_s^{\text{int}}, \quad k_0 = \frac{2\pi}{\lambda_s}, \quad k'_s = \frac{2\pi n_s}{\lambda_s}. \quad (2)$$

Both parametrizations describe the same κ_s ; the external one carries no n_s .

Deriving Eq. (9) from the appendix (external angle)

Carry (1) to (λ_s, θ_s) via

$$(S1) \quad d\omega_s = \frac{2\pi c}{\lambda_s^2} d\lambda_s \quad \left(\omega_s = \frac{2\pi c}{\lambda_s} \right), \quad (3)$$

$$(S2) \quad \kappa_s d\kappa_s = \frac{1}{2} k_0^2 \sin(2\theta_s) d\theta_s = \frac{2\pi^2}{\lambda_s^2} \sin(2\theta_s) d\theta_s \quad (\kappa_s = k_0 \sin \theta_s), \quad (4)$$

with $\omega_s \omega_i = 4\pi^2 c^2 / (\lambda_s \lambda_i)$ inside A_0 and the signal azimuth $d\phi$ kept as a differential. The numeric coefficient is

$$\frac{1}{8\pi^4} \cdot 4\pi^2 \cdot 2\pi \cdot 2\pi^2 = 2\pi, \quad (5)$$

so

$$N_s(\lambda_s, \theta_s, \phi) = 2\pi \sqrt{2\pi} \frac{\hbar d_{\text{eff}}^2 \omega_p L^2 N_p}{\varepsilon_0 n_s n_i n_p \lambda_s^5 \lambda_i} \sin(2\theta_s) d\lambda_s d\theta_s d\phi \int d\xi_i e^{-\frac{1}{2}(\xi_s - \xi_i)^2} \text{sinc}^2\left(\frac{1}{2}L \Delta k_z\right) \quad (6)$$

which is the printed Eq. (9) *verbatim*, apart from the $d\phi$. The printed equation is therefore a flux density per unit azimuth; the full ring carries an extra factor $\int_0^{2\pi} d\phi = 2\pi$. This is not a numerical error in the paper: Sec. III.D states explicitly that the flux is integrated over " $\phi = 0 \rightarrow 2\pi$ ", so the 2π enters at the numerical-integration stage. The only defect of Eq. (9) is the missing $d\phi$ symbol.

Here $\xi_s = \kappa_s W$ and $\Delta k_z = \sqrt{k_s'^2 - \kappa_s^2} + \sqrt{k_i'^2 - (\xi_i/W)^2} - k_p'$ (appendix form; the in-medium wavenumbers stay primed). Integrating (6) over the azimuth reproduces $\eta = 2N_s/N_p = 2.72 \times 10^{-10}$ and $2N_s = 4.44 \times 10^7 \text{ s}^{-1}$ ($\theta_m = 29.12^\circ$, $P_p = 80 \text{ mW}$), matching the paper's Table I. Implemented in `spdc_physics.py` (`Ns` with `external=True`; the equivalent internal-angle parametrization $\kappa_s = k'_s \sin \theta'_s$, whose Jacobian carries $\frac{1}{2} k_s'^2 = \frac{1}{2} n_s^2 k_0^2$, is available as `external=False` and yields the identical total flux).

Where the old " $2\pi n_s^2$ " claim came from. Reading " $\kappa_s = k_s \sin \theta_s$ " with the *in-medium* $k_s = 2\pi n_s / \lambda_s$ (and hence θ_s internal) makes the Jacobian (S2) acquire a factor n_s^2 , and folding the azimuthal 2π into the prefactor as well makes the printed Eq. (9) appear short by $2\pi n_s^2 = 2\pi(1.660)^2 \approx 17.3$. Under Fig. 2's convention neither factor is missing: the n_s^2 never belonged there, and the 2π is the azimuth integral the authors perform explicitly.

The genuine misprints: Eq. (7)

The factor of 4 ($2\pi^4$ vs $8\pi^4$). Eq. (7) and (1) are the same equation up to the relabeling $\int d^2\xi \rightarrow \int d^2\xi_i$ with $\xi = (\kappa_s + \kappa_i)W$, $\xi_i = \kappa_i W$. At fixed κ_s this is a pure *shift*,

$$\int d^2\xi f(\xi) = \int d^2\xi_i f(\xi_s + \xi_i), \quad \text{Jacobian} = 1, \quad (7)$$

which cannot alter the prefactor, so one of the two printed constants is a misprint. The appendix's $8\pi^4$ is the correct one, on three independent grounds: (i) it reproduces the paper's simulated η (Table I: 2.6×10^{-10} ; our integral: 2.72×10^{-10}); (ii) it agrees with the closed-form Eq. (8) — taken from the independent Ref. [18] (Koch et al., IEEE J. Quantum Electron. **31**, 769 (1995)) — which evaluated directly gives $\eta = 2.63 \times 10^{-10}$; (iii) it matches the measured $\eta \approx 2.7 \times 10^{-10}$ to 2.8×10^{-10} . With $2\pi^4$ everything would sit $4\times$ above all three.

The exponent sign. Eq. (7) prints $e^{+\frac{1}{2}\xi^2}$, which diverges; the appendix's $e^{-\frac{1}{2}|\xi_s+\xi_i|^2}$ is correct.

Summary.

Item	Location	Status
$e^{+\frac{1}{2}\xi^2}$	Eq. (7) exponent	misprint; must be $-$ (appendix has it right)
$2\pi^4$	Eq. (7) prefactor	misprint; correct value $8\pi^4$ (appendix, Eq. (8), experiment)
missing $d\phi$	Eq. (9)	notation only; Eq. (9) is a per-unit-azimuth density
“dropped $2\pi n_s^2$ ”	Eq. (9)	retracted: θ_s is the external angle (Fig. 2), so the printed Jacobian coefficient is exact